TP1 Altergrad

1.) We can work on different aspect in order to improve our basic self-attention mechanism such as the **representation of the sentence** [1], **add more attention layers**, and have a **penalization term** added to the loss function at the end that prohibits redundancies in the result attention matrix.
Indeed, since we would like to get more precision we want to get more variable that could capture more data in order to create a better understanding of context: how the words together are connected in terms of meaning, in terms of their function in the sentence, hence one vector to embed a whole phrase would probably have the tendency to lose a bit of information. We would like to use more information than just a one vector. Hence we could represent our sentence as a $2D$ - $l_0 \times N$ matrix with N the number of words include in our sentence, and l_0 the total number of tokens used in our tokenization display. [2]. So the ith column of our matrix corresponds to the encoded vector of the $i$the word of the sentence. this idea we ll get back to it using [9] afterwards at question (2).
So this gives us more possibilities to understand connections between words in order to capture a more precise understanding of the general sentence that can be ground for the attention layer.
Also, we can increase the number of attention layers. Each attention layer captures a certain function in our sentence. For instance, one attention layer could focus more on the connections some adjectives share with a noun giving a better representation of the noun or object described. So when we increase the number of attention layers in parallel we increase the number of functions at stake that allow us to understand a sentence in depth.
As the multi attention layers are in parallel, we could expect that during the results from each layers could share some redundancies. In that matter, we could add a **penalization term** on the loss function at the end on our annotation matrix A[3] which forces each vector of the matrix to be focused on a single aspect.
On a technical point of view, we could improve our attention mechanism which is more focused on the semantics by adding a mechanism into the layers which entails a special focus on the relations between words. We could indeed use a LSTM[4] on each hidden layers: once a calculation has been made we get the result in the hidden layer $(h_1, h_2, \ldots, h_n)$ in order to **target more lexical correlations between one word and its predecessor (first direction) or the word and its successor (second direction)**. Combining the two gives us a new vector $\tilde{h}_i$ for each i.

[1] we say one sentence as the group of word given in input. It could actually be more than one sentence grammatically speaking or even a part of a sentence
[2] for instance, Open ai uses on gpt2 50 thousands tokens to encode words or strings
[3] We would add $||AA^T - I||^2_{Frobenius}$ to the loss function
[4] or in that case biLSTM (both ways)

2.) There are many motivations for replacing recurrent operations with self-attention which are: **ability to capture long range distance dependencies**, **parallelization** in computing usage, **not subject to information loss**.
Based on [9] "attention is all you need", the self attention captures long term dependencies between words: RNN's are not very good at memorizing on a long time (due to issues such as the gradient vanishing problem) so if words are spaced a lot we loose dependencies we could have capture. Self attention mechanism does'nt do that because it has direct access to tokens accross long distance.
[9] is a strong advocate of multi head attention layers that can be parallelized. So functioning on a GPU, provides faster computations during training and usage.
Also, as we are not using RNN's in self attention we are not using memory for the hidden states. Hidden states are used and given to the newt step thanks to compression: we loose information from step n to step n+1.

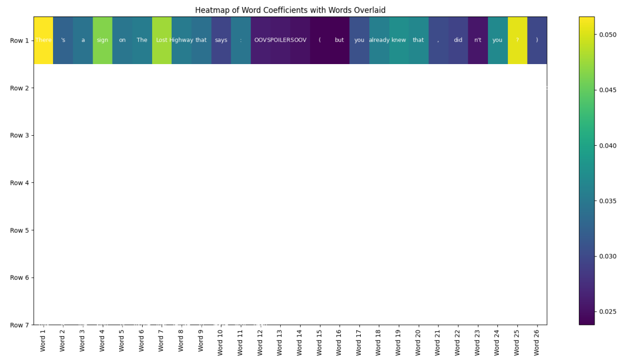

Figure 1: Caption

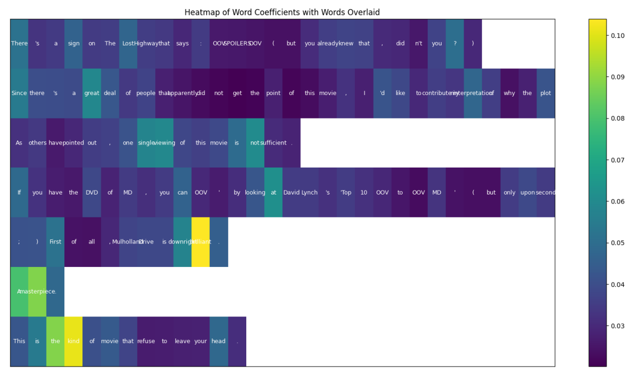

Figure 2: Example on the paragraph

3.)

(If you can't see the picture on the docuent please look at them in the folder, thanks) We took the paragraph provided in the example. We can see that the heatmap show us high values for adjectives meliorative such as great nice etc... and extreme when detected the word "masterpiece" for instance. So it shows that it can truly focus on meaningful positive parts of positive comments and highlight them. The model works as it should !

4.) In the Han architecture each sentences are coded independently from one to the other. This fact means that the model isn't exploiting the relationships between sentences in the document. It implies that the HAN architecture doesn't benefit from the advantage of context as context in a document is rarely compacted in one sentence but is however really adding a lot to understand for instance nuances, the objects the document is describing etc...
Specific to our model, we could probably improve it by adding attention layers. Even if this can help to use a bit more context by parallelizing attention layers it may not fully use the whole strength of context in the reading of a document.